

1. Importing required Libraries

```
In [1]: from bs4 import BeautifulSoup
import requests
import pandas as pd
import numpy as np
```

2. Defining Functions to transform the data

2.1 Function to extract Product Title

```
In [2]: def get_title(soup):

    try:
        # Outer Tag Object
        title = soup.find("span", attrs={"id":'productTitle'})

        # Inner NavigatableString Object
        title_value = title.text

        # Title as a string value
        title_string = title_value.strip()

    except AttributeError:
        title_string = ""

    return title_string
```

2.2 Function to extract Product Price

```
In [3]: def get_price(soup):

    try:
        price = soup.find("span", attrs={'id':'a-price aok-align-center reinventPricePriceToPayMargin priceToPay'}).string.strip()

    except AttributeError:

        try:
            # If there is some deal price
            price = soup.find("span", attrs={'id':'a-size-large a-color-price savingPriceOverride aok-align-center reinventPriceSavingsPercentageMargin savingsPercentage'}).string.strip()

        except:
            price = ""

    return price
```

2.3 Function to extract Product Rating

```
In [4]: def get_rating(soup):

    try:
        rating = soup.find("i", attrs={'class':'a-icon a-icon-star a-star-4-5'}).string.strip()

    except AttributeError:

        try:
            rating = soup.find("span", attrs={'class':'a-icon-alt'}).string.strip()

        except:
            rating = ""

    return rating
```

2.4 Function to extract Number of User Reviews

```
In [5]: def get_review_count(soup):

    try:
        review_count = soup.find("span", attrs={'id':'acrCustomerReviewText'}).string.strip()

    except AttributeError:
        review_count = ""

    return review_count
```

2.5 Function to extract Availability Status

```
In [6]: def get_availability(soup):

    try:
        available = soup.find("div", attrs={'id':'availability'})
        available = available.find("span").string.strip()

    except AttributeError:
        available = "Not Available"

    return available
```

3. Rearrange the data into CSV Format

```
In [7]: if __name__ == '__main__':

    # to tell the website that we are not a bot we need to pass headers (using user-agent)
    HEADERS = {
        "User-Agent": "your user agent",
        "Accept-Language": "en-US, en;q=0.5",
    }

    # url from searching "playstation 5" in amazon.in
    URL = "https://www.amazon.in/s?k=playstation+5&crd=3P5HNQ43SNCIU&prefix=playstation+5%2Caps%2C233&ref=nb_sb_noss_1"

    # making HTTP request to the website
    webpage = requests.get(URL, headers=HEADERS, timeout=10)

    # Soup Object containing all data
    soup = BeautifulSoup(webpage.content, "html.parser")

    # Fetch links as List of Tag Objects
    links = soup.find_all("a", attrs={'class':'a-link-normal s-underline-text s-underline-link-text s-link-style a-text-normal'})

    # Store the links
    links_list = []

    # Loop for extracting links from Tag Objects
    for link in links:
        links_list.append(link.get('href'))

    d = {"title":[], "price":[], "rating":[], "reviews":[], "availability":[]}

    # Loop for extracting product details from each link
    for link in links_list:
        new_webpage = requests.get("https://www.amazon.in" + link, headers=HEADERS)

        new_soup = BeautifulSoup(new_webpage.content, "html.parser")

        # Function calls to display all necessary product information
        d['title'].append(get_title(new_soup))
        d['price'].append(get_price(new_soup))
        d['rating'].append(get_rating(new_soup))
        d['reviews'].append(get_review_count(new_soup))
        d['availability'].append(get_availability(new_soup))

    # Now we will export the data into "Amazon Data.csv" file
    amazon_df = pd.DataFrame.from_dict(d)
    amazon_df['title'].replace('', np.nan, inplace=True)
    amazon_df = amazon_df.dropna(subset=['title'])
    amazon_df.to_csv("Amazon Data.csv", header=True, index=False)
```

4. View the saved Data Frame

```
In [8]: amazon_df
```

```
Out[8]:
```

	title	price	rating	reviews	availability
0	Sony PS5 PlayStation Console		4.3 out of 5 stars	555 ratings	In stock
1	Sony PS5 PlayStation Console+God Of War Ragnar...		4.0 out of 5 stars	13 ratings	In stock.
2	Sony God Of War Ragnarok Standard Edition ...		4.7 out of 5 stars	2,823 ratings	In stock
3	Hogwarts Legacy Standard Edition PlayStatio...		4.7 out of 5 stars	93 ratings	In stock
4	Square Enix Final Fantasy Xvi, Standard Editio...		Previous page	This item will be released on June 22, 2023.	
5	PlayStation 5 DualSense Wireless Controller		4.7 out of 5 stars	38,702 ratings	Only 1 left in stock
6	Far Cry 6 - PlayStation 5 Standard Edition		4.3 out of 5 stars	19 ratings	In stock
7	Cult of the lamb Standard Edition PlayStat...		Previous page		In stock
8	Call of Duty: Vanguard (PS5) - PlayStation 5		4.3 out of 5 stars	25 ratings	Only 2 left in stock
9	Sony Horizon Forbidden West Standard Edition...		4.6 out of 5 stars	303 ratings	In stock
10	Dorca® Dust Cover for Sony Playstation 5 Cover...		4.5 out of 5 stars	417 ratings	In stock
11	WALLTON - PS5 Skin Protective Wrap Cover Vinyl...		2.9 out of 5 stars	5 ratings	In stock
12	PS5 Grand Theft Auto V		4.4 out of 5 stars	238 ratings	In stock
13	2K GAMES Ps5 Wwe 2K22		4.3 out of 5 stars	98 ratings	In stock
14	Electronic Arts Dead Space Standard Edition ...		4.8 out of 5 stars	7 ratings	In stock
15	GLITCH ps5 console carry bag bags for trav...		Previous page		In stock

16	Benazcap for PS5 Faceplate, Hard Shockproof Pl...	4.7 out of 5 stars	3,309 ratings	
17	Capcom Street Fighter 6, Standard Edition, Pla...	1.0 out of 5 stars	1 rating	Only 1 left in stock